

Task 10. Use Matplotlib module for plotting in python

Date: 24/9/2

Aim: To use matplotlib module for plotting in python.

problem 10.1: write a python programming to display a bar of the popularity of programming languages.

Algorithm:

1. Define two lists for programming language and their popularity respectively
2. find the maximum popularity value in the list
3. Define a scaling factor to scale the bar heights within a certain limit (eg. 50 characters)
4. for each language and popularity pairs, calculate the bar height as the popularity value scaled by scaling factor.
5. print the chart using a loop to iterate over the programming language list: a. print the language name. b. use a loop to print the bar chart by printing the bar character (eg. "*") a number of times equal to the bar height c. print the popularity value with a separator character d. print a newline character.

Program:

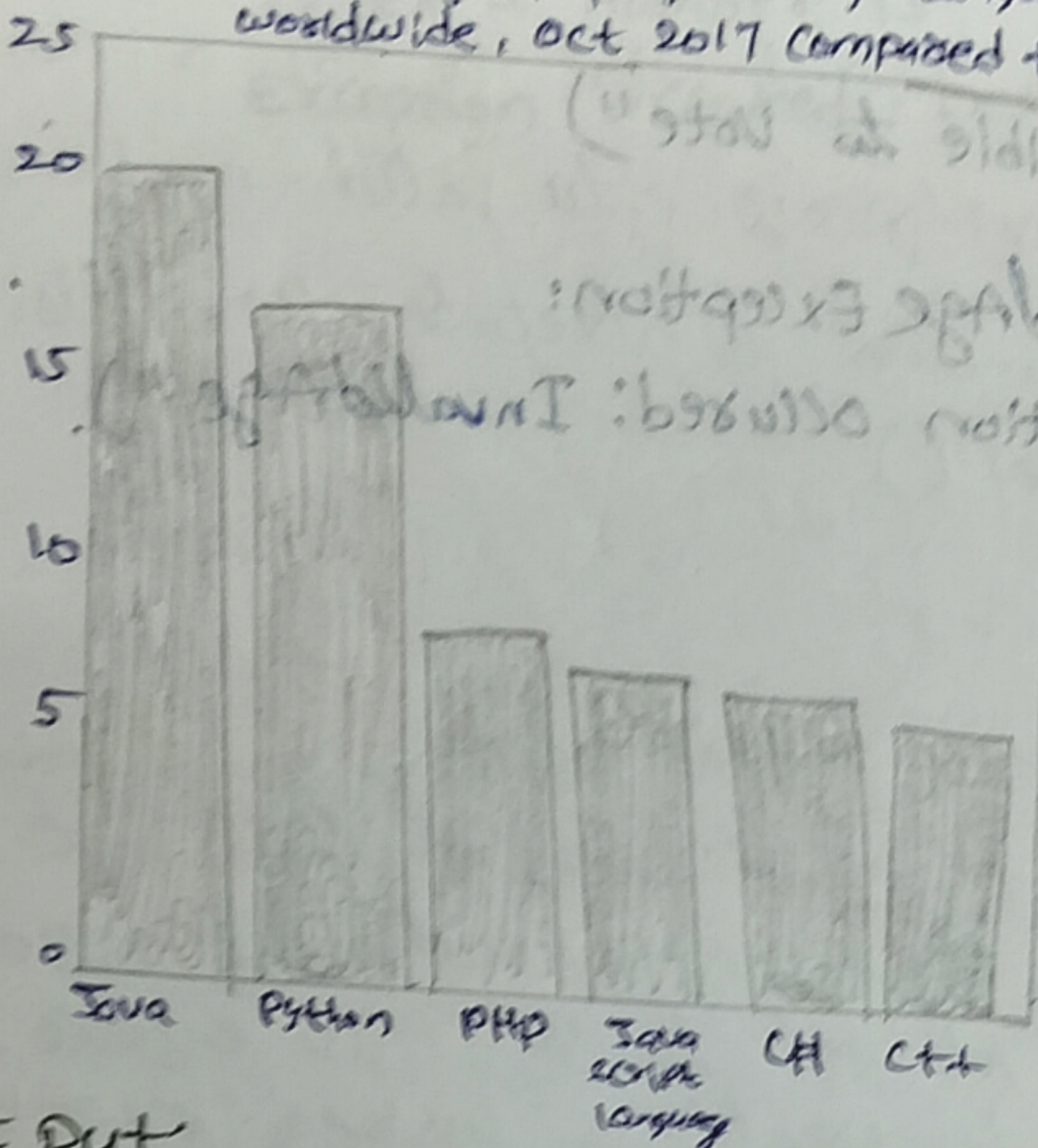
```
#pip install matplotlib
import matplotlib.pyplot as plt

languages = ['Java', 'python', 'PHP', 'JavaScript', 'C#', 'C++']
Popularity = [22.2, 17.6, 8.8, 8.7, 7.7, 6.7]

plt.bar(languages, Popularity, color='b')
plt.title('popularity of programming languages')
plt.xlabel('programming languages')
plt.ylabel('popularity')
plt.show()
```

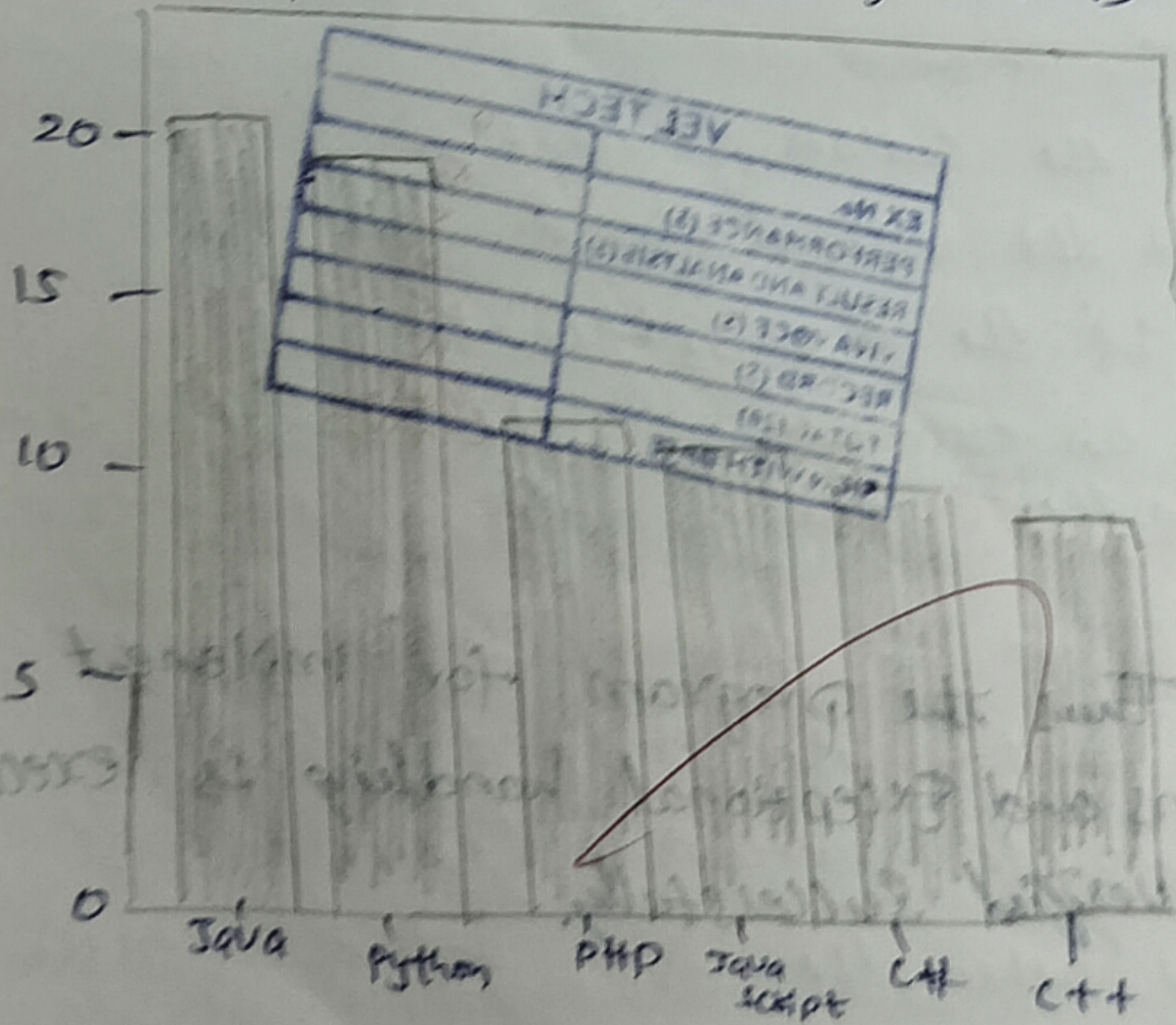

Sample data:
 Programming languages: Java, Python, PHP, Javascript, C#, C++
 Popularity: 22.2, 17.6, 8.8, 7.7, 6.7

Sample out:
 Popularity of programming language worldwide, Oct 2017 compared to a year ago



Out put

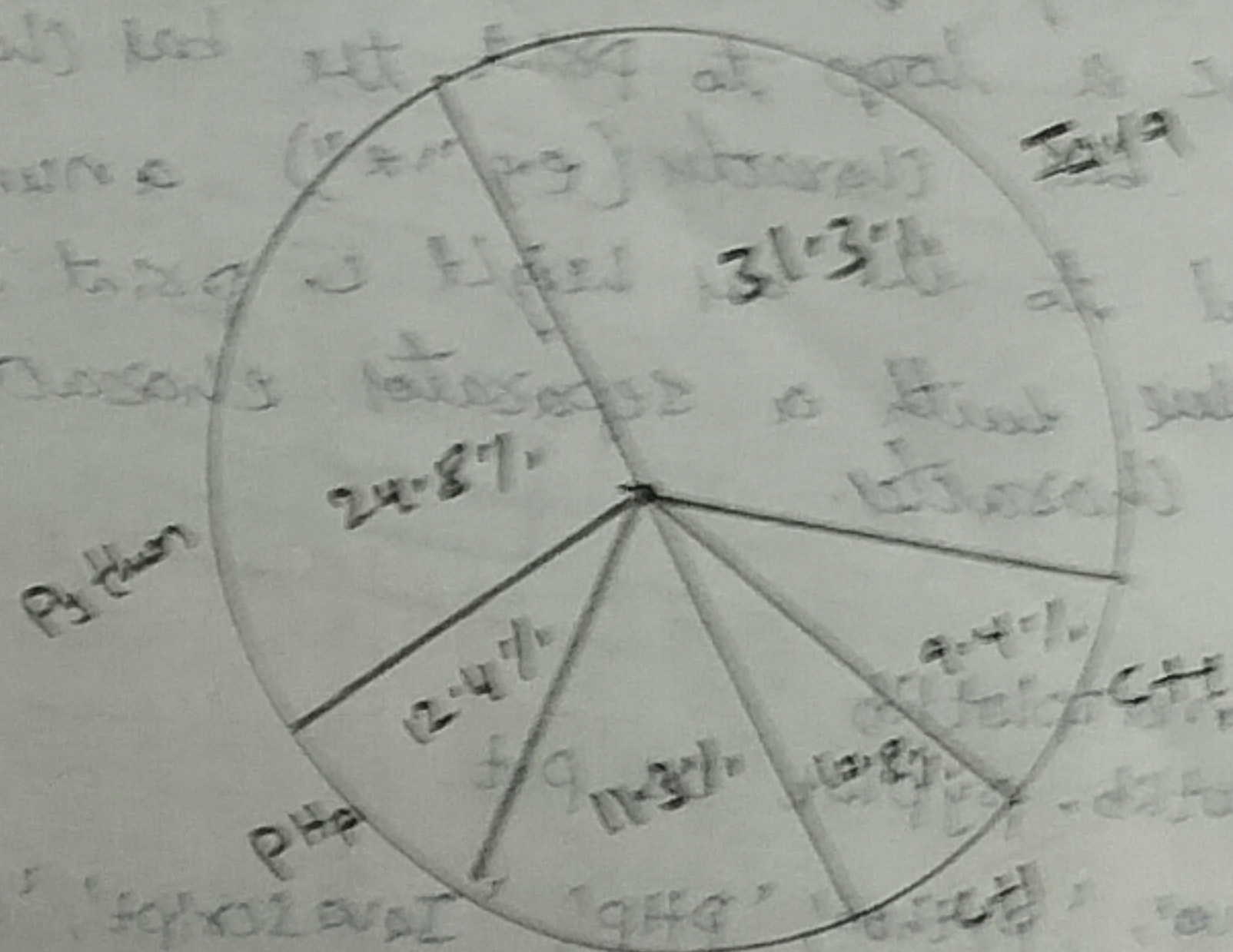
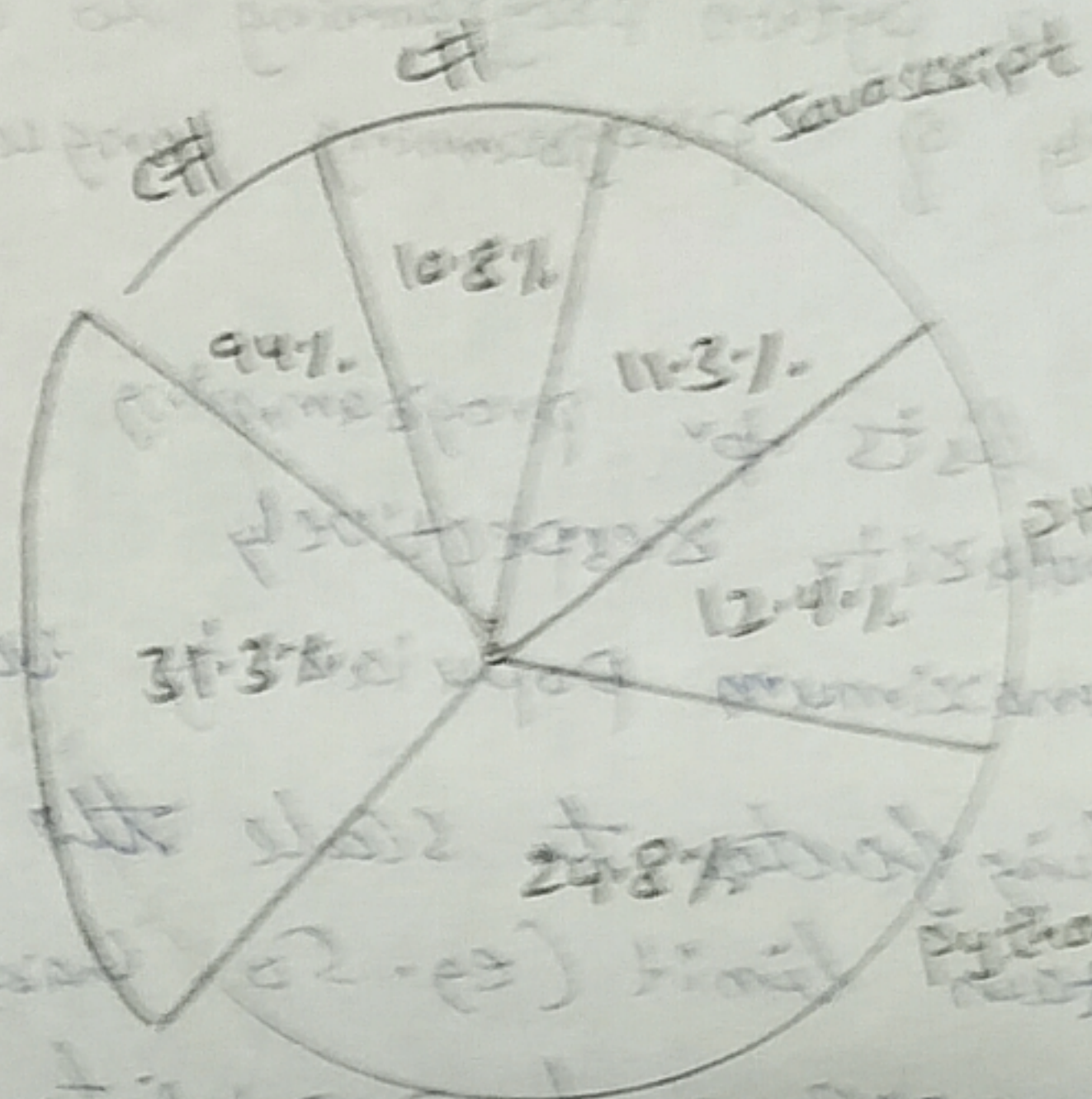
Popularity of programming language



programming languages

✓

Sample output:-



problem 10.2 write a python programming to create a pie chart of the popularity of programming languages.

Sample data: programming languages: Java, python, PHP, Javascript, C#, C++

Popularity: 22.2, 17.6, 8.8, 8, 7.7, 6.7

Algorithm:-

1. create a list of programming languages and popularity
2. create a pie chart using the matplotlib library,
3. set the title and legend for the pie chart
4. show the pie chart.

Program:-

```
import matplotlib.pyplot as plt
```

step 1

```
languages = ['Java', 'python', 'PHP', 'Javascript', 'C#', 'C++']  
popularity = [22.2, 17.6, 8.8, 8, 7.7, 6.7]
```

step 2

```
plt.title('Popularity', labels = languages, autopct = '%1.1f%%')
```

step 3

```
plt.title('popularity of programming languages')  
plt.legend(languages, loc = "best")
```

step 4

```
plt.show()
```

VELTECH	
EX No.	10
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	5
TOTAL (20)	15
SIGN WITH DATE	8/10

Result:- Thus the python program use matplotlib module for plotting is executed and verified successful.